MSG No. 63.459 $Cm'cm$ [Type III, orthorhombic]

Table 1: Wyckoff site: `4a`, site symmetry: `2/m'..`

No.	position	mapping
1	$[0, 0, 0]$	`[1,2,7,8]`
2	$\left[0, 0, \frac{1}{2}\right]$	`[3,4,5,6]`
3	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[9,10,15,16]`
4	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[11,12,13,14]`

Table 2: Wyckoff site: `4b`, site symmetry: `2/m'..`

No.	position	mapping
1	$\left[0, \frac{1}{2}, 0\right]$	`[1,2,7,8]`
2	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[3,4,5,6]`
3	$\left[\frac{1}{2}, 0, 0\right]$	`[9,10,15,16]`
4	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[11,12,13,14]`

Table 3: Wyckoff site: `4c`, site symmetry: `m'2'm`

No.	position	mapping
1	$\left[0, y, \frac{1}{4}\right]$	`[1,4,5,8]`
2	$\left[0, -y, \frac{3}{4}\right]$	`[2,3,6,7]`
3	$\left[\frac{1}{2}, y+\frac{1}{2}, \frac{1}{4}\right]$	`[9,12,13,16]`
4	$\left[\frac{1}{2}, \frac{1}{2}-y, \frac{3}{4}\right]$	`[10,11,14,15]`

Table 4: Wyckoff site: `8d`, site symmetry: `-1'`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, 0\right]$	`[1,15]`
2	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[2,16]`
3	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[3,13]`
4	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[4,14]`
5	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[5,11]`
6	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[6,12]`
7	$\left[\frac{3}{4}, \frac{3}{4}, 0\right]$	`[7,9]`
8	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[8,10]`

Table 5: Wyckoff site: `8e`, site symmetry: `2..`

No.	position	mapping
1	$[x, 0, 0]$	`[1,2]`
2	$\left[x, 0, \frac{1}{2}\right]$	`[3,4]`
3	$\left[-x, 0, \frac{1}{2}\right]$	`[5,6]`
4	$[-x, 0, 0]$	`[7,8]`

continued ...

Table 5

No.	position	mapping
5	$\left[x+\frac{1}{2},\ \frac{1}{2},\ 0\right]$	[9,10]
6	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	[11,12]
7	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	[13,14]
8	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ 0\right]$	[15,16]

Table 6: Wyckoff site: 8f, site symmetry: m'..

No.	position	mapping
1	$[0,\ y,\ z]$	[1,8]
2	$[0,\ -y,\ -z]$	[2,7]
3	$\left[0,\ -y,\ z+\frac{1}{2}\right]$	[3,6]
4	$\left[0,\ y,\ \frac{1}{2}-z\right]$	[4,5]
5	$\left[\frac{1}{2},\ y+\frac{1}{2},\ z\right]$	[9,16]
6	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ -z\right]$	[10,15]
7	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[11,14]
8	$\left[\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	[12,13]

Table 7: Wyckoff site: 8g, site symmetry: ..m

No.	position	mapping
1	$\left[x,\ y,\ \frac{1}{4}\right]$	[1,4]
2	$\left[x,\ -y,\ \frac{3}{4}\right]$	[2,3]
3	$\left[-x,\ y,\ \frac{1}{4}\right]$	[5,8]
4	$\left[-x,\ -y,\ \frac{3}{4}\right]$	[6,7]
5	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{4}\right]$	[9,12]
6	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{3}{4}\right]$	[10,11]
7	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ \frac{1}{4}\right]$	[13,16]
8	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ \frac{3}{4}\right]$	[14,15]

Table 8: Wyckoff site: 16h, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$[x,\ -y,\ -z]$	[2]
3	$\left[x,\ -y,\ z+\frac{1}{2}\right]$	[3]
4	$\left[x,\ y,\ \frac{1}{2}-z\right]$	[4]
5	$\left[-x,\ y,\ \frac{1}{2}-z\right]$	[5]
6	$\left[-x,\ -y,\ z+\frac{1}{2}\right]$	[6]
7	$[-x,\ -y,\ -z]$	[7]
8	$[-x,\ y,\ z]$	[8]
9	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z\right]$	[9]
10	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ -z\right]$	[10]
11	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[11]

continued ...

Table 8

No.	position	mapping
12	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$	[12]
13	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$	[13]
14	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	[14]
15	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, -z\right]$	[15]
16	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, z\right]$	[16]